

RESULTS

Results

01 · INDEPENDENT T-TEST

do two groups' means differ?

Table 1. Group statistics

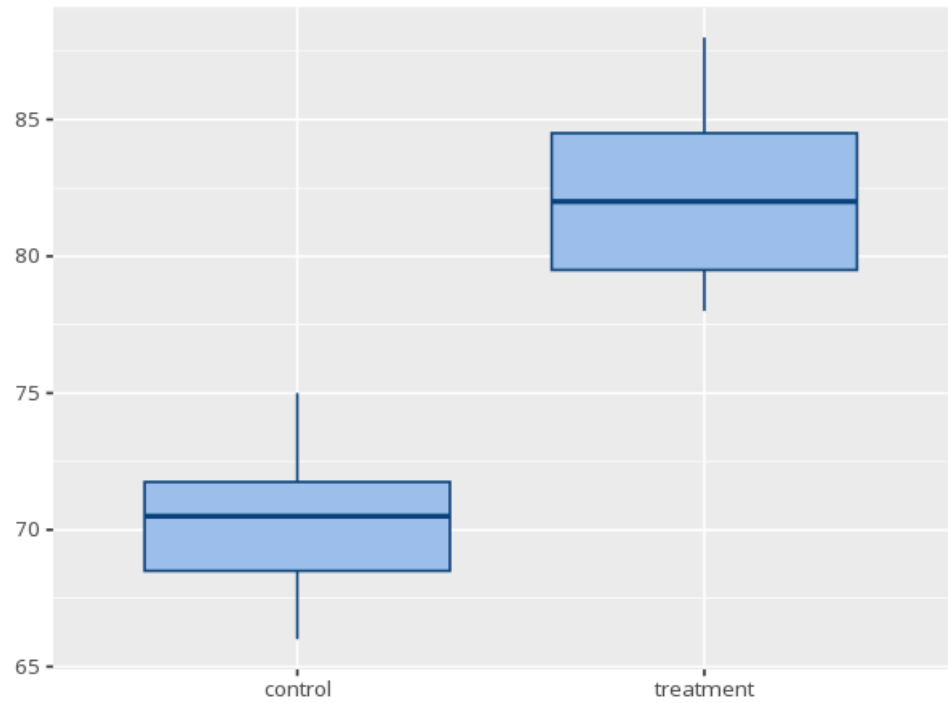
Group	N	Mean	Std. Dev.	SE
control	6	70.33	3.14	1.28
treatment	6	82.33	3.78	1.54

Table 2. Independent-samples t-test

Contrast	t	df	p	M _{diff}	95% CI	Cohen's d [95% CI]
control – treatment	-5.98	9.68	<.001	-12.00	[-16.49, -7.51]	-3.45 [-5.32, -1.53]

assumption checks: Levene's test for equal variances & within-group normality (Shapiro-Wilk per group); a Welch row replaces the pooled row when equal-variance is off. (Levene F=0.38, p=.550 · Shapiro control W=0.99, p=.995; Shapiro treatment W=0.96, p=.847 · Welch test) — equal variances look reasonable — normality looks reasonable in both groups

Figure. Distribution of the outcome by group



Understanding the numbers

t. How many standard errors the two group means are apart. Here, $t(9.68) = -5.98$.

p. The probability of a difference this large (or larger) if the groups truly had equal means. Here, $p < .001$ - below the conventional .05 threshold, a significant difference.

Cohen's d. The size of the gap in standard-deviation units (~0.2 small, 0.5 medium, 0.8 large). Here, $d = -3.45 [-5.32, -1.53]$.

N. The number of complete cases (rows) in each group. Here, $n = 6$ for control and $n = 6$ for treatment.

Mean. The average outcome value in each group. Here, $M = 70.33$ for control and $M = 82.33$ for treatment.

SD. How spread out the outcome values are around each group's mean. Here, $SD = 3.14$ for control and $SD = 3.78$ for treatment.

SE. The precision of each group's mean as an estimate of its population mean. Here, $SE = 1.28$ for control and $SE = 1.54$ for treatment.

df. The degrees of freedom for the t-test (pooled or Welch-adjusted). Here, $df = 9.68$.

M diff. The difference between the two group means. Here, the mean difference is -12.00 .

95% CI. The range of mean differences plausible at the chosen confidence level, given this sample. Here, the 95% CI for the mean difference is $[-16.49, -7.51]$.

How to read this test

Asks whether two groups' averages differ by more than chance. Look at p: below alpha (e.g. 0.05) means significant. The mean difference and 95% CI show the size and precision of the gap; Cohen's d gives effect size (~0.2 small, 0.5 medium, 0.8 large). A large p means no significant difference was detected — not that the group means are equal.

APA template: An independent-samples t-test compared control ($M=70.3$, $SD=3.1$) and treatment ($M=82.3$, $SD=3.8$), $t(9.68)=-5.98$, $p < .001$, $d=-3.45 [-5.32, -1.53]$.

Statistical basis: Independent-samples t-test (pooled variance) (Student (1908). "The probable error of a mean." *Biometrika*, 6(1), 1-25.); Welch's correction (unequal variances, the app's default) (Welch, B. L. (1947). "The generalization of Student's problem when several different population variances are involved." *Biometrika*, 34(1-2), 28-35.); Cohen's d effect-size benchmarks (.2/.5/.8) (Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Routledge.); Levene's test for equal variances (Brown-Forsythe median-centered variant) (Levene, H. (1960). "Robust tests for equality of variances." In I. Olkin (Ed.), *Contributions to Probability and Statistics* (pp. 278-292). Stanford University Press.); Brown-Forsythe median-centered variant of the homogeneity-of-variance test (Brown, M. B., Forsythe, A. B. (1974). "Robust tests for the equality of variances." *Journal of the American Statistical Association*, 69(346), 364-367.). Full references in the exported CITATIONS.txt.

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